Lab Activity: LaTeX Fundamentals

September 3, 2025

Lab Activity – Assignment (20 points)

Use only the LaTeX commands shown in the Lab Guide.

Q1. Document Basics and Math Integration (5 points)

1. Create a title page with your name, roll number, and today's date.
2. Write a short paragraph on the topic "Technology in Scientific Writing". In your paragraph, include one emphasized word, one proper quotation, and at least one escaped special character.
3. Write the below three variables inline: one with a subscript, one with a superscript, and one as a Greek letter.

   p_1, q^2, μ
4. Typeset the below formula for the sum of the first n integers.

   $\sum_{k=1}^{n} \frac{k}{n}$
5. Make a bulleted list of any three advantages of structured writing systems.

Q2. Advanced Features (5 points)

1. Write the n-th term of a geometric progression (GP) using both subscripts and superscripts.

   $a_n = a_1 r^{n-1}$
2. Typeset the below exponential equation :-

$$k = Ae^{-\frac{E_a}{RT}}$$

3. Insert a figure into the document, scaled to 40% of text width, and add a caption below it.

4. Write the below displayed equation :-

$$S_n = \frac{n}{2}(a_1 + a_n)$$

5. Write the below mathematical statement.

$$a^2 + b^2 \geq 2ab$$

Q3. Replicate the Csp203.pdf as it is. (10 points)